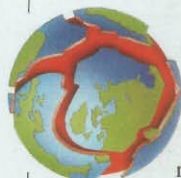


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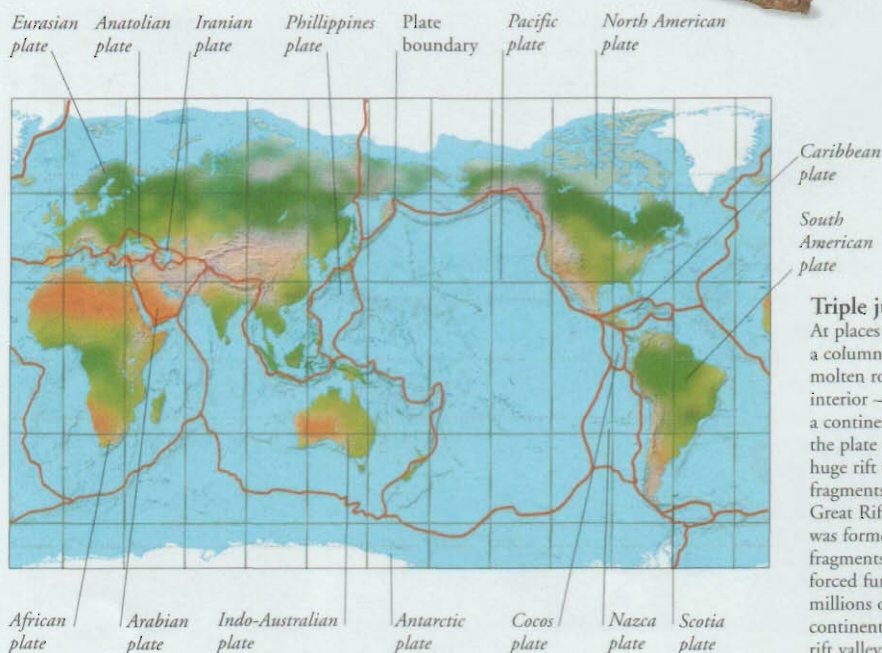
THE WORLD'S SEVEN great land masses are known as continents. The seven continents are North America, South America, Africa, Europe, Asia, Antarctica, and

Australia. Although you may not realize it, the continents are crunching together and drifting apart even as you read this page. This is because the Earth's outer shell, or crust, is made up of a number of vast, ever-moving slabs of rock called tectonic plates. The continents are embedded in these plates, which move very slowly – Europe and North America, for example, are drifting apart by just 4 cm (1.5 in) each year. Over millions of years, however, the continents have shifted this way and that across the globe, dramatically changing the face of the planet time and time again.



Tectonic plates

There are nine major tectonic plates and a number of smaller ones. They fit together like the pieces of a jigsaw, covering the whole of the Earth's surface. The continents are carried by continental plates, such as the Eurasian plate. Oceanic plates, such as the Pacific plate, form most of the seafloor; the rest is made up of the fringes of the continental plates, which lie underwater.



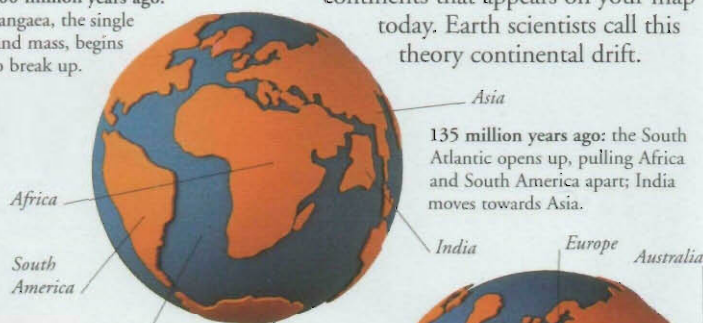
The major plates of the Earth's crust

Continental drift

If you study a world map, you will notice that the east coast of South America looks as if it would fit neatly into Africa's west coast. This is because 220 million years ago Africa, South America, and all the other continents formed a single giant "supercontinent", now called Pangaea. Eventually, Pangaea split up into smaller land masses that drifted over the Earth's surface, giving the arrangement of continents that appears on your map today. Earth scientists call this theory continental drift.



200 million years ago: Pangaea, the single land mass, begins to break up.



135 million years ago: the South Atlantic opens up, pulling Africa and South America apart; India moves towards Asia.

10 million years ago: Antarctica and Australia drift apart; the North Atlantic opens up, moving North America away from Europe.

Glossopteris fossil



Fossilized *Lystrosaurus* skull



Antarctica

Evidence for continental drift

Identical fossils of land-based plants and animals, such as the fern *Glossopteris* and the mammal *Lystrosaurus*, have been found in continents now widely separated by the sea. The only plausible explanation is that the continents were once linked together.

Alfred Wegener

German meteorologist and geophysicist Alfred Wegener (1880–1930) devised the theory of continental drift. As evidence, he cited the continents' matching coastlines, similar rock strata in continents separated by huge oceans, and fossil discoveries. Although widely accepted now, his ideas were ridiculed at the time.



Triple junctions

At places called triple junctions, a column of magma – hot, molten rock from the Earth's interior – burns its way through a continental plate. This splits the plate three ways, producing huge rift valleys between the fragments of the plate. The Great Rift Valley in East Africa was formed in this way. The fragments of the plate are forced further apart over millions of years, creating new continental land masses. As the rift valleys widen, new oceans form between the pieces of the fragmented plate.



Satellite image of Africa's Great Rift Valley